

Data Wrangling Coding Questions with Sales Data

Data Context

You have access to the file `sales_data_raw.csv`. It is a real-world inspired sales dataset for an electronics and accessories distributor with the following columns:

- OrderID, Date, Region, Product, Category, Quantity, UnitPrice, CustomerID, SalesRep, PaymentMethod, TotalAmount
- The dataset includes missing values, duplicate records, inconsistent text capitalization/spaces, and quantity outliers.

Coding Questions

1. Load and Inspect the Data

- a) Read the CSV into a Pandas DataFrame and display the first 5 rows.
- b) Display the number of missing values for each column.

2. Missing Values

- a) Fill missing `Region` and `SalesRep` values with the string "Unknown".
- b) For missing `PaymentMethod`, fill with the mode method.

3. Duplicates

- a) Count the number of duplicate rows in the dataset.
- b) Drop complete duplicates and reset the DataFrame index.

4. Text Cleaning and Standardization

- a) Remove leading/trailing spaces and set the `'Product'` names to proper case (e.g., "Phone").
- b) Standardize `'Region'` to title case (e.g., "North").

5. Outlier Identification & Treatment

- a) Calculate mean and standard deviation of `Quantity`. Identify rows where `Quantity` is more than 3 std deviations above the mean.
- b) Replace these outlier `Quantity` values with the median `Quantity` for their `'Product'` group.

6. Feature Engineering

- a) Extract the month and year from the Date column into new columns.
- b) Flag orders larger than the 95th percentile of TotalAmount with a boolean 'HighValue' column.

7. Aggregations & Grouping

- a) Compute total and average revenue per Region and Product combination.
- b) For each month, show the best-selling product by total quantity.

8. Pivot and Reshape

- a) Create a pivot table with Region as rows, Product as columns, and sum of TotalAmount as values.

9. SalesRep Analytics

- a) Which SalesRep handled the highest revenue overall? Show their total revenue.
- b) List the top 3 SalesReps by total orders handled each.

10. Export the Cleaned Data

- Save the fully cleaned and engineered DataFrame to sales_data_cleaned.csv.

Solutions (Code)

1

```
import pandas as pd
df = pd.read_csv('sales_data_raw.csv')
print(df.head())
print(df.isnull().sum())
```

2

```
df['Region'].fillna('Unknown', inplace=True)
df['SalesRep'].fillna('Unknown', inplace=True)
payment_mode = df['PaymentMethod'].mode()[0]
df['PaymentMethod'].fillna(payment_mode, inplace=True)
```

3

```
print(df.duplicated().sum())
df = df.drop_duplicates().reset_index(drop=True)
```

4

```
df['Product'] = df['Product'].str.strip().str.title()
df['Region'] = df['Region'].str.strip().str.title()
```

5

```
mean_q = df['Quantity'].mean()
std_q = df['Quantity'].std()
outlier_mask = df['Quantity'] > (mean_q + 3 * std_q)
product_medians = df.groupby('Product')['Quantity'].transform('median')
df.loc[outlier_mask, 'Quantity'] = product_medians[outlier_mask]
```

6

```
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.month
df['Year'] = df['Date'].dt.year
hi_val = df['TotalAmount'] > df['TotalAmount'].quantile(0.95)
df['HighValue'] = hi_val
```

7

```
grouped = df.groupby(['Region', 'Product'])['TotalAmount'].agg(['sum', 'mean'])
monthly_best = df.groupby(['Year', 'Month', 'Product'])['Quantity'].sum().reset_index()
monthly_best = monthly_best.loc[monthly_best.groupby(['Year', 'Month'])['Quantity'].idxmax()]
```

8

```
pivot = pd.pivot_table(df, index='Region', columns='Product', values='TotalAmount', aggfun
```

9

```
salesrep_revenue = df.groupby('SalesRep')['TotalAmount'].sum()
top_salesrep = salesrep_revenue.idxmax()
print(top_salesrep, salesrep_revenue[top_salesrep])
top3_orders = df.groupby('SalesRep')['OrderID'].count().nlargest(3)
print(top3_orders)
```

10

```
df.to_csv('sales_data_cleaned.csv', index=False)
```

End of Questionnaire